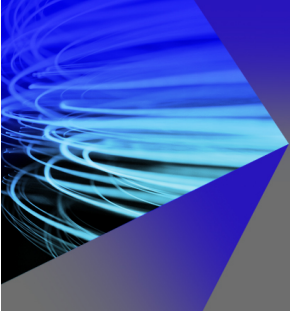


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Unveiling Auger recombination current component in rectification curves of silicon heterojunction solar cells

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ABSTRACT

The rectification characteristics of semiconductor junctions are the basis for their application in diverse electronic and optoelectronic devices. Theoretically, the manifestation of Auger recombination under high positive bias voltage enables solar cells to achieve a high fill factor (FF); however, other recombination mechanisms and high series resistance make it difficult to be observed in the rectification curve. Here, the Auger recombination current component is identified in the rectification curves of silicon heterojunction solar cells featuring high FF and high open-circuit voltage (Voc). The current-voltage (I-V) characteristics of the device with high-conductivity nanocrystalline silicon oxide emitter demonstrate a distinct diode component exhibiting an ideality factor of $n = 2/3$, which is a definitive indicator of Auger recombination. The temperature-dependent I-V characteristics and Suns-Voc measurements reveal that suppression of series resistance and defect-mediated recombination is essential for unveiling the Auger recombination component.

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The recorded efficiencies of various silicon solar cells have been improved continuously in recent years.^{1–3} Among the different solar cells, silicon heterojunction (SHJ) solar cells exhibit high open-circuit voltage (Voc) and fill factor (FF), with a record efficiency exceeding 27%.² Many other high-efficiency SHJ solar cells with a FF of over 86% have been reported recently.^{4,5} In ideal cases, the rectification curve (current-voltage curve) of a SHJ solar cell follows the Shockley equation, which is dominated by the Shockley diffusion mechanism with an ideality factor $n = 1$. However, in real devices, series resistance and additional recombination pathways—such as Shockley-Read-Hall (SRH) recombination in the depletion region and Auger recombination in the high injection condition—can significantly alter its I-V characteristics. Auger recombination limits the solar cell efficiency by introducing a three-carrier nonlinear recombination pathway in high injection cases (under a high forward bias or illuminated by strong light), and provides a diode component with an ideality factor $n = 2/3$ in the equivalent circuit of the solar cell.⁶ According to the

expression for FF proposed by Green,⁷ for a SHJ cell with an open-circuit voltage of 750 mV,³ the maximum fill factor (FFm) is predicted to be 85.4% with an ideality factor $n = 1$ at 300 K, and it is 89.3% with $n = 2/3$. That is to say, although Auger recombination reduces the cell efficiency, it can lead to a higher fill factor theoretically. When both surface/bulk defective recombination and series resistance are sufficiently suppressed, the Auger recombination effect is uncovered.^{8,9} Therefore, many theoretical analyses based on the light current-voltage (I-V) curve and numerical simulations reveal that high FF of SHJ solar cells could be related to Auger recombination.^{6,8,9} Auger recombination also has a significant impact on the performance of many other semiconductor devices. For example, it can lead to a decline in the emission efficiency of bipolar junction transistors (BJTs),¹⁰ and an efficiency droop in light-emitting diodes (LEDs).^{11,12} However, direct experimental observation of Auger recombination through the rectification characteristics of the various diode devices remains elusive, as its contribution is often overshadowed by other

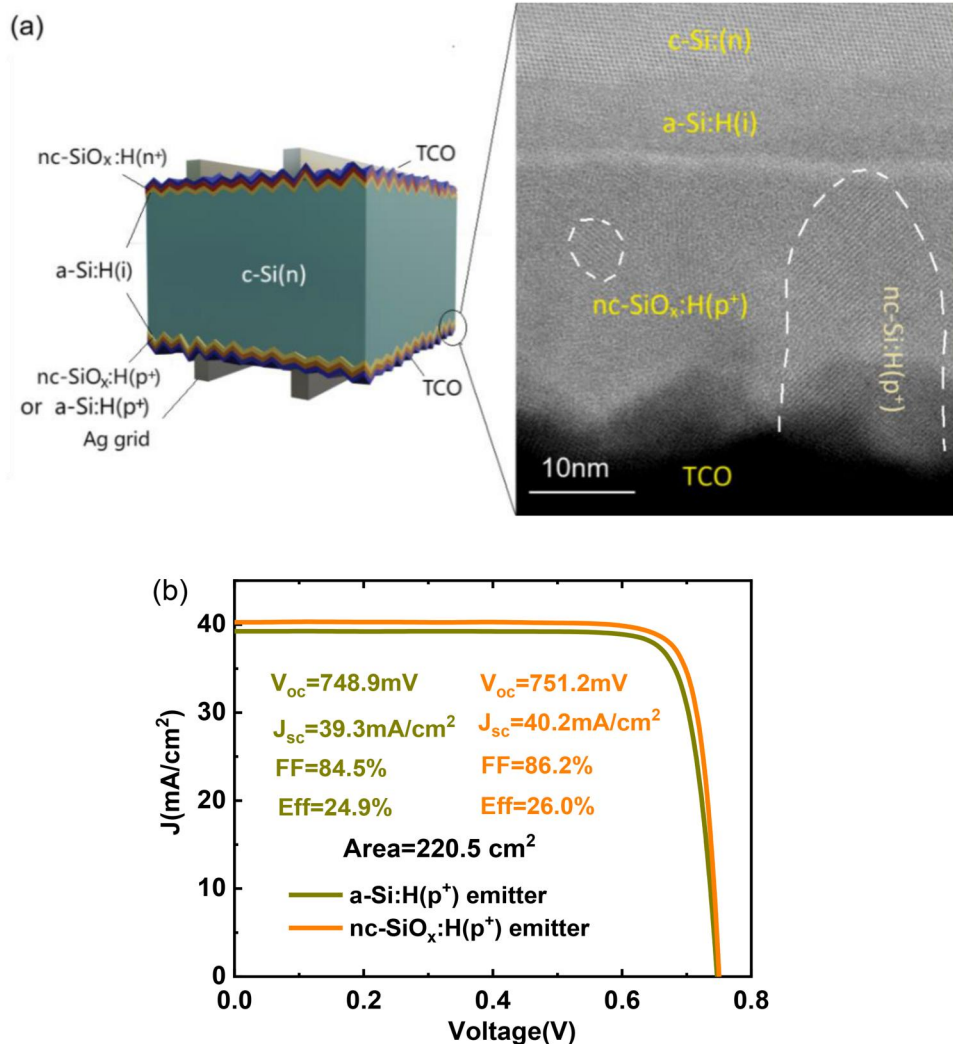


FIG. 1. (a) Structure diagram of SHJ solar cell and cross-sectional AC-TEM image of a SHJ solar cell with $\text{nc-SiO}_x\text{:H}(\text{p}^+)$ emitter. (b) Light J - V curves of SHJ solar cells with $\text{nc-SiO}_x\text{:H}(\text{p}^+)$ and $\text{a-Si:H}(\text{p}^+)$ emitters.

recombination mechanisms and series resistance effect. Temperature-dependent dark I - V analysis is a powerful diagnostic tool in exploring the carrier transport mechanisms of various homojunction and heterojunction semiconductor devices. Here, we identified the distinct signature of Auger recombination current in the I - V characteristics of the high efficiency SHJ solar cells.

The structure diagram of SHJ solar cells with a p-type nanocrystalline silicon oxide emitter [$\text{nc-SiO}_x\text{:H}(\text{p}^+)$] or amorphous silicon [$\text{a-Si:H}(\text{p}^+)$] emitter is illustrated in Fig. 1(a). The front-field and back-junction configuration was employed. From top to bottom, the layers consist of silver grids, transparent conductive oxide (TCO) film (105 nm), n-type hydrogenated nanocrystalline silicon oxide [$\text{nc-SiO}_x\text{:H}(\text{n}^+)$] layer (30 nm), intrinsic amorphous silicon [$\text{a-Si:H}(\text{i})$] passivation layer (7 nm), crystalline silicon (c-Si) wafer (130 μm), $\text{a-Si:H}(\text{i})$, p-type $\text{nc-SiO}_x\text{:H}(\text{p}^+)$ (25 nm) or $\text{a-Si:H}(\text{p}^+)$ (13 nm) emitter, TCO, and silver grids, respectively (experimental details in the supplementary material). The light current density-voltage (J - V) curves of the SHJ solar cells with $\text{nc-SiO}_x\text{:H}(\text{p}^+)$ and $\text{a-Si:H}(\text{p}^+)$ emitters are

presented in Fig. 1(b). The photovoltaic parameter is provided in the supplementary material (Table S1). The solar cell with a $\text{nc-SiO}_x\text{:H}(\text{p}^+)$ emitter demonstrated superior performance with higher V_{oc} , short-circuit current density (J_{sc}), and FF compared to the solar cell with an $\text{a-Si:H}(\text{p}^+)$ emitter, consequently achieving a high efficiency of 26%, in contrast to the latter's 24.9% efficiency. A remarkable FF of 86.2% is realized for the solar cell featuring $\text{nc-SiO}_x\text{:H}(\text{p}^+)$ emitter, whereas the device with a conventional $\text{a-Si:H}(\text{p}^+)$ emitter exhibits a lower FF of 84.5%.

The performance difference between the two solar cells primarily originates from the distinct structural characteristics of the emitters. The spherical aberration-corrected transmission electron microscope (AC-TEM) image of a SHJ solar cell with a $\text{nc-SiO}_x\text{:H}(\text{p}^+)$ emitter is given in Fig. 1(a). It can be seen that the $\text{nc-SiO}_x\text{:H}(\text{p}^+)$ emitter consists of both nanocrystalline silicon ($\text{nc-Si:H}(\text{p}^+)$) and amorphous silicon oxide [$\text{a-SiO}_x\text{:H}(\text{p}^+)$] phases. Some large $\text{nc-Si:H}(\text{p}^+)$ grains span the entire emitter, establishing direct conductive bridges between the $\text{a-Si:H}(\text{i})$ passivation layer and TCO electrode. Consequently,

photogenerated carriers preferentially migrate through the highly conductive nc-Si:H(p⁺) grain pathways,⁵ leading to the transport mechanisms in the c-Si(n)/a-Si:H(i)/nc-SiOx:H(p⁺) contact that more closely resemble those of an interface-passivated crystalline silicon homojunction. This behavior should differ from the well-documented carrier transport characteristics of a conventional heterojunction based on the c-Si(n)/a-Si:H(i)/a-Si:H(p⁺) structure.^{13,14}

To understand the difference between the two kinds of SHJ solar cells with different emitters, especially to reveal the correlation between FF and transport mechanisms, small-area (1.5 cm²) SHJ solar cells with nc-SiOx:H(p⁺) and a-Si:H(p⁺) emitters were fabricated for dark I-V characterization. The small-area solar cells maintain the core device architecture depicted in Fig. 1(a) but feature a chemically polished surface instead of a textured one and use thermally evaporated silver electrodes to replace screen-printed silver grids on both sides. The four-electrode method was employed for dark I-V measurements (see the supplementary material, Fig. S1).

Figures 2(a) and 2(b) are, respectively, the dark I-V curves (solid dots) of the devices with a-Si:H(p⁺) and nc-SiOx:H(p⁺) emitters at 400 K. For the device featuring an a-Si:H(p⁺) emitter, the experimental I-V characteristics were analyzed using a widely adopted double-diode model^{15,16} as depicted in the inset of Fig. 2(a). As a conventional analytical framework for characterizing the behavior of various diodes, the double-diode model can be qualitatively described as

$$I = \left(\sum I_j \right) + I_{sh}, \quad I_{sh} = \frac{V - IR_s}{R_{sh}}, \quad (1)$$

$$I_j = I_{0j} \left\{ \exp \left[\frac{q(V - IR_s)}{n_j kT} \right] - 1 \right\},$$

where I_{sh} is the parallel current, R_s and R_{sh} are the series and parallel resistance, I_{0j} and n_j are the reverse saturation current and ideality factor of the diode j , q is the elementary charge, k is the Boltzmann constant, and T is the absolute temperature. For a typical silicon-based pn junction, $j = 2$. I_1 is the Shockley diffusion (SD) current with $n_1 = 1$, and I_2 is the generation/recombination (G/R) current with $n_2 = 2$.¹⁷ Transport mechanisms of the SHJ solar cells additionally involve thermionic emission (TE) and multitunneling capture

emission (MTCE).¹⁷ The dark I-V curve of the device with a-Si:H(p⁺) emitter can be well fitted by the double-diode model (green solid line) with $n_1 = 1.14$, $n_2 = 3.5$, and $R_s = 110$ mΩ. Under a high forward bias condition (>0.6 V), the experimental I-V curve shows convex bending from the exponential relationship and exhibits a linear trend. The I-V characteristics (>0.6 V) without considering series resistance are represented by the red dotted line in Fig. 2(a). Therefore, this deviation is commonly attributed to the series resistance in semiconductor physics textbooks and literature.^{10,18,19}

For the device with a nc-SiOx:H(p⁺) emitter [Fig. 2(b)], unexpectedly, the experimental I-V curve exhibits concave upward curvature in semi-log coordinates, deviating from the double-diode model fit (green dashed line; $n_1 = 1.14$, $n_2 = 2.8$, $R_s = 1$ mΩ) at forward bias exceeding 0.6 V, persisting even when R_s was set to be zero. Therefore, the three-diode model [inset in Fig. 2(b)] was employed. The corresponding current expression for the three-diode model is similar to Formula (1), except that j can also take on a value of 3. When R_s and the ideality factor n_3 of the third diode are, respectively, set to be 1 mΩ and 0.67 (blue dashed line), the experimental I-V curve can be well fitted (violet solid line). The third diode with $n_3 = 0.67$ should correspond to the Auger recombination mechanism whose contribution to the current is manifested under high bias voltage. By comparison, for the device with the a-Si:H(p⁺) emitter [Fig. 1(a)], the high R_s induces a linear I-V relationship under high injection, thereby covering up the influence of Auger recombination in the measured I-V characteristics. The I-V characteristics of previously reported diverse diodes with high series resistance could be well fitted using the double-diode model.^{15,16} Regarding the SHJ solar cell incorporating highly conductive ultra-thin p-type and n-type nc-SiOx:H layers as emitter and front contact layers, along with optimized interface contact resistances,⁵ the total series resistance is significantly reduced, which enables distinct identification of Auger recombination contributions in the dark I-V characteristics. Under this condition, the three-diode model becomes essential for accurately fitting the rectification characteristics.

To gain deeper insight into the transport mechanisms, the dark I-V curves of the device with a nc-SiOx:H(p⁺) emitter at positive bias voltage in the temperature range of 260–440 K were further measured

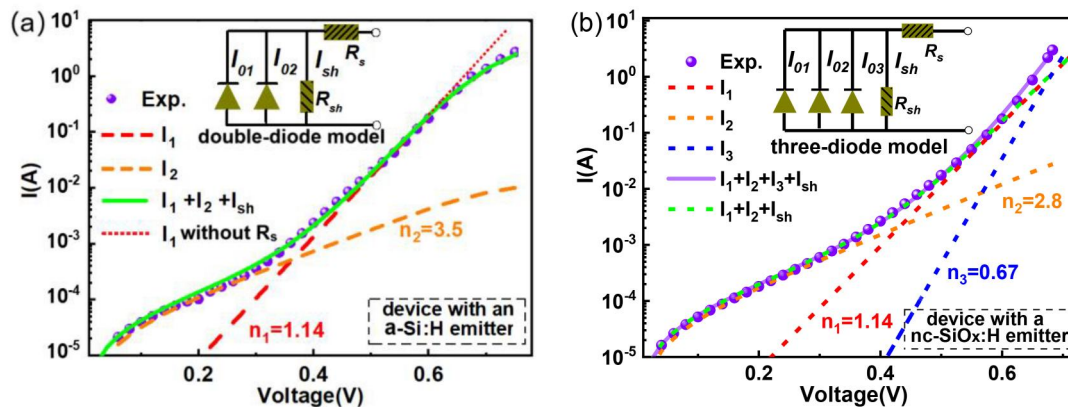


FIG. 2. (a) and (b) Experimental and fitting dark I-V curves of the devices with a-Si:H(p⁺) and nc-SiOx:H(p⁺) emitters at 400 K under forward bias. The insets in (a) and (b) are the double- and three-diode models for fitting the I-V curves, respectively.

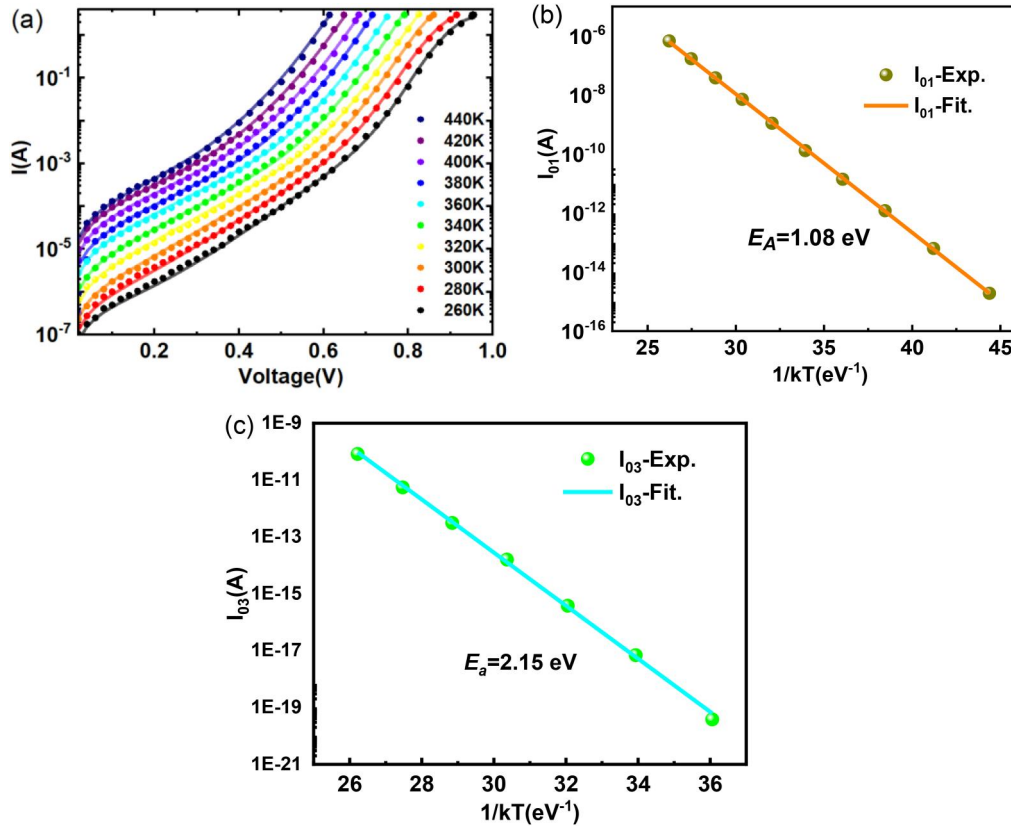


FIG. 3. Analysis of dark I-V curves of a SHJ device with nc-SiOx:H(p $^{+}$) emitter. (a) Measured (dots) and fitted (solid lines) dark I-V curves in the temperature range of 260–440 K for $V > 0$ V. (b) The linear relationship between I_{01} and $1/(kT)$ in a semi-log plot. (c) $I_{03} \sim 1/(kT)$ relationship in semi-log plot.

and analyzed as shown in Fig. 3. The results under negative bias voltage are given in the [supplementary material](#) (Fig. 2S). The three-diode model with ideality factors $n_1 = 1.14$, $n_2 = 2.8$, and $n_3 = 0.67$ was employed to fit the set of I-V curves. The remaining fitting parameters, including the saturation currents (I_{01} , I_{02} , I_{03}), series resistance (R_s), and shunt resistance (R_{sh}), are summarized in the [supplementary material](#) (Table S2), where the fitting details and error analysis are also provided. At high temperatures (>340 K), the experimental I-V curves exhibit a superlinear growth in semi-log plots under high bias, demonstrating the signature of Auger recombination current. At low temperatures (<300 K), the curves transition to a sublinear behavior due to increase in the series resistance, which could be attributed to the increase in contact resistivity at the TCO/nc-SiOx:H(p $^{+}$) interface.^{20,21} Combined with the fitting parameters (in the [supplementary material](#)), we conclude that minimizing the series resistance below 1 m Ω enables the observation of Auger recombination current in the rectification curve. Temperature dependence of the saturation current I_{01} can be fitted according to the formula $I_{01} = I_{001} \exp(-EA/kT)$,^{10,17} where I_{001} is the pre-factor and EA the activation energy. The fitted $I_{01} \sim (1/kT)$ relationship in semi-log plot is shown in Fig. 3(b). The extracted activation energy EA of 1.08 eV is close to the bandgap of c-Si at 360 K (1.10 eV), which indicates that SD mechanism dominates in the medium voltage region of the I-V characteristics.^{10,17} The ideality factor n_2 , which is higher than 2 indicates

that the MTCE mechanism could still exist in the SHJ solar cell with a nc-SiOx:H emitter due to the presence of amorphous phases.

The current density of Auger recombination for high-level injection (with high forward bias) can be expressed as^{22,23}

$$I_3 I_{03} \exp \left\{ \frac{q(V - IR_s)}{n_3 kT} \right\}, \quad n_3 = \frac{2}{3}, \quad I_{03} = A q n_i^3 W (C_n + C_p). \quad (2)$$

Here, C_n and C_p are the Auger recombination coefficient for electrons and holes, respectively. W and q are the thickness of bulk and elementary charge, respectively. n_i is the intrinsic carrier density, and n_i^3 indicates that Auger recombination is a process involving three carriers. Since $n_i^2 \propto T^3 \exp(-E_g/kT)$,²¹ $n_i^3 \propto T^{4.5} \exp(-1.5E_g/kT)$ and $I_{03} \propto \exp(-1.5E_g/kT)$. Fitting the $I_{03} \sim (1/kT)$ curve gives the active energy E_a of 2.15 eV as shown in Fig. 3(c), which is larger than $1.5E_g$ (if $E_g = 1.10$ eV). This may be due to measurement errors or limitations of the fitting method. For example, the Auger recombination coefficient is treated as a temperature-independent constant here. However, experimental observations indicate that it increases with decreasing temperature.²⁴

The ideality factor n under high injection conditions can also be extracted from the Suns-Voc method, which inherently excludes the influence of the series resistance. According to the expression of V_{oc} given by²⁵

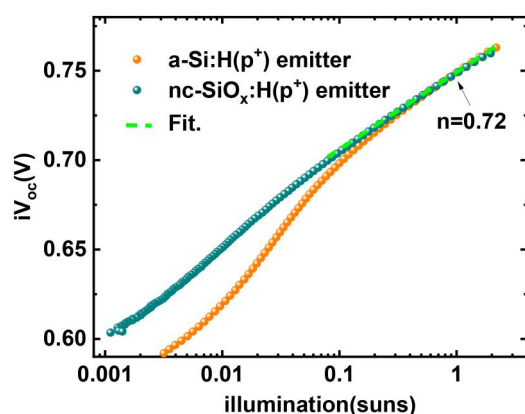


FIG. 4. Illumination-dependent implied V_{oc} characteristics of SHJ solar cells incorporating nc-SiO_x:H(p⁺) and conventional a-Si:H(p⁺) emitters. The unit of illumination is standard sun. 1 sun = 1000 (W/m²).

$$V_{oc} = \frac{nkT}{q} \ln \left(\frac{J_{sc}}{J_0} \right), \quad (3)$$

where J_0 is the saturate current density and J_{sc} is proportional to illumination (ϕ), the implied V_{oc} (iV_{oc}) vs $\ln(\phi)$ can be obtained through Suns-Voc measurements. Fitting the $iV_{oc} \sim \ln(\phi)$ curve linearly, the ideality factor n can be determined from the slope. The illumination dependence of iV_{oc} for the devices with nc-SiO_x:H(p⁺) and a-Si:H(p⁺) emitters is shown in Fig. 4 (measured by Sinton WCT-120). Under high injection conditions (>1 Sun illumination), the $iV_{oc} \sim \ln(\phi)$ curves of both devices demonstrate identical slopes of 0.0188 V, yielding an experimental ideality factor of 0.72, which closely matches the theoretical ideality factor of 2/3 expected for the Auger recombination condition. Benefiting from the excellent passivation capability of intrinsic a-Si:H layers¹ and superior bulk quality of crystalline silicon substrates, both the investigated SHJ solar cells exhibit remarkably high minority carrier lifetimes (>3 ms). In this case, the SRH recombination related to bulk and surface defects is sufficiently suppressed in the SHJ solar cells despite the emitter difference. Moreover, as mentioned above, The Suns-Voc measurement eliminates the influence of series resistance. Therefore, the effects of Auger recombination become identified in the $iV_{oc} \sim \ln(\phi)$ curve under high injection conditions for both SHJ devices.

Based on the dark I-V and Suns-Voc measurements, the Auger recombination current component is observed in the rectification curves of high-efficiency SHJ solar cells with the high-conductivity nc-SiO_x:H(p⁺) emitter and high-quality intrinsic a-Si:H passivation layer. The factors that affect Auger recombination, such as series resistance and defect-related recombination, have been clearly understood. The ability to distinguish and quantify Auger recombination in dark I-V measurements offers a powerful tool for advanced device diagnostics and material engineering in photovoltaics.

See the [supplementary material](#) for experimental details of large- and small-area SHJ solar cells, the fitting method of the dark J-V curve, fitting curves under reverse bias as well as fitting parameters of the J-V curve at different temperatures.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Junming Huang: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Writing – original draft (equal). **Dongshen Liu:** Investigation (equal); Software (equal); Validation (equal); Writing – original draft (equal). **Fengzhen Liu:** Conceptualization (equal); Funding acquisition (equal); Methodology (equal); Resources (equal); Visualization (equal); Writing – review & editing (equal). **Wanwu Guo:** Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal). **Tianyu Li:** Data curation (equal); Investigation (equal); Validation (equal). **Yuqin Zhou:** Conceptualization (equal); Data curation (equal); Methodology (equal). **Yurong Zhou:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Project administration (equal); Resources (equal).

DATA AVAILABILITY

The data that support the findings of this study are available within the article and its [supplementary material](#).

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